HingSent

Sentiment Analysis of Hinglish E-Commerce Reviews

A fine-tuned **HingBERT** model for three-class sentiment classification (Positive, Negative, Neutral) on **Hinglish e-commerce reviews**.

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Resources

- **Hugging Face Model** — Arman-03/sentiment-analysis-hinglish
- **Live Demo (Streamlit)** — hingsent.streamlit.app
- **GitHub Repository** — github.com/Arman-03/HingSent

Problem

Indian e-commerce reviews are mostly written in Hinglish (Hindi + English mixed text).

Standard English or Hindi models struggle due to:

- Vocabulary mismatch
- Code-mixing at word/phrase/sentence level
- Sentiment ambiguity in mixed reviews

Example Reviews

Review	Label
"Zabardast product hai! Delivery bhi super fast thi, paise vasool ekdum."	Positive
"Bilkul bakwaas quality, waste of money bhai. Return karna padha."	Negative
"Theek thaak hai product. Na bahut achha na bura, average quality milti hai."	Neutral

Dataset

- 1,048 Hinglish e-commerce reviews
- Generated using Gemini 1.5 Flash
- Labeled using Gemini 1.5 Flash
- 3 classes: Positive, Negative, Neutral
- Deduplicated to remove template artefacts
- Split (80/10/10):

Train	Validation	Test
838	105	105

Our Approach

We fine-tune **HingBERT** (`13cube-pune/hing-bert`) on Hinglish reviews for three-class sentiment classification.

Hinglish Review → HingBERT (Pre-trained) → Fine-tuned Classifier → Positive / Negative / Neutral

- **Tokenizer:** HingBERT tokenizer (51,200 tokens)
- **Model:** 12 layers, 768 hidden size, 12 attention heads
- **Training:** 5 epochs, batch size 16, AdamW optimizer
- **Loss:** Cross Entropy Loss

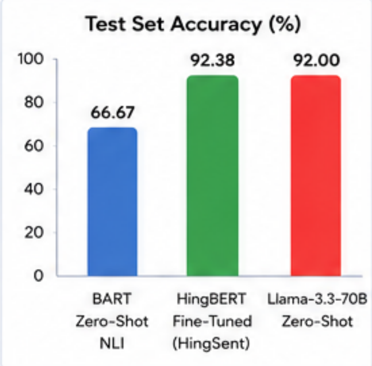
Results

Model Comparison

Model	Accuracy	Macro F1
BART Zero-Shot NLI (Baseline)	66.67%	0.657
HingBERT Fine-Tuned **(HingSent)**	92.38%	0.923
Llama-3.3-70B Zero-Shot (Upper Bound)	92.00%	0.920

✓ HingSent outperforms the BART baseline by a large margin.

✓ HingSent matches the 70B Llama-3.3 model while using far fewer parameters.



Features (Web Application)

- Single review sentiment prediction
- Batch CSV processing
- Confidence visualization
- Word clouds and sentiment analytics
- Downloadable labeled results
- Deployed on Hugging Face Spaces

Impact

Enables e-commerce platforms to:

- Automatically analyze customer sentiment
- Detect negative trends early
- Improve product quality and customer experience
- Understand Indian consumer behavior at scale

Domain-specific fine-tuning beats scale alone for code-mixed NLP.

Tech Stack

HingBERT | Transformers | PyTorch | Streamlit | Hugging Face | Groq API | Python